

ZHIYU NI

Email: zhiyuni@berkeley.edu Website: <https://zyni2001.github.io>

EDUCATION

University of California, Berkeley (UC Berkeley) PhD student, <i>Computer Science</i> , Advisor: Pierluigi Nuzzo	<i>Berkeley, CA</i> 2025.1 - Fall 2027 (expected)
University of Southern California (USC) PhD student, <i>Electrical Engineering</i>	<i>Los Angeles, CA</i> 2022.8 - 2024.12
University of Science and Technology of China (USTC) B.S. <i>Physics</i> , Outstanding Graduates (top 3%)	<i>Hefei, China</i>

PUBLICATIONS

Analyzing Robustness of Graph Lottery Tickets (**ICASSP 2024 Oral**)

Finding Robust Graph Lottery Tickets (**TMLR 2024**)

Let me grok for you: Accelerating grokking via embedding transfer from a weaker model (**ICLR 2025**)

Efficient On-Device Video Out-of-Distribution Detection (**ICCV 2025**)

Defenses Against Prompt Attacks Learn Surface Heuristics (**ACL 2026**)

Agentified Assessment of Logical Reasoning Agents (**ICLR 2026 agents in the wild Workshop**)

Improving the Reliability of Auto-formalization for Logical Reasoning (**Arxiv**)

ACIES: A Self-Evolving Agentic Contract Framework for Verified Controller Synthesis in Cyber-Physical Systems (REALM 2026 Workshop, co-located with EMNLP 2026)

WORK EXPERIENCES

Research Scientist Intern (Meta)	<i>Sunnyvale, CA</i> 2026.5 - Now
- Developing a domain foundation model for ads ranking. - Proposed length-aware sequence modeling to adaptively learn user behavior, delivering a 0.04% normalized entropy (NE) gain.	
AI Research Intern (Synopsys DSO.AI)	<i>Sunnyvale, CA</i> 2025.5 - 2025.8
- Developed Analog Agent to automate function and structure identification from circuits. - Created a hybrid pipeline integrating Graph Neural Networks with subgraph isomorphism test, achieving 100% non-overlapping block recovery.	
NLP Research Intern (iFLYTEK)	<i>Hefei, China</i> 2022.2 - 2022.6
- Trained large-scale multi-layer transformers for a bidirectional machine translation task. - Achieved 15% higher BLEU score compared to Google Translate.	

RESEARCH EXPERIENCE

Graph Lottery Tickets: Robustness Analysis and Adversarial Sparsification (ICASSP 2024 Oral) (TMLR 2024)

- Analyzed the vulnerability of sparse graph lottery tickets to structural attacks, then developed ARGS to jointly prune perturbed graph edges and GNN weights using label- and homophily-aware objectives.
- Produced highly sparse GNNs with substantially stronger adversarial robustness while reducing inference cost across multiple graph benchmarks.

Accelerating Grokking via Embedding Transfer from a Weaker Model (ICLR 2025)

- Co-developed GrokTransfer, which initializes a target model with embeddings learned by a weaker model; proved direct generalization on synthetic XOR and eliminated delayed generalization in neural networks and Transformers.
- Validated the method across diverse tasks and architectures, demonstrating more predictable and computationally efficient generalization dynamics.

Secure On-Device Video Out-of-Distribution Detection (ICCV 2025)

- Developed SecDOOD, a secure cloud-device framework combining hypernetwork-based personalization with selective feature encryption to match full fine-tuning performance without device-side backpropagation.
- Designed dynamic feature sampling to encrypt only informative channels, reducing encryption overhead while preserving personalized OOD detection performance.

Defenses Against Prompt Attacks Learn Surface Heuristics (ACL 2026)

- Built controlled diagnostics revealing position, token-trigger, and topic-generalization shortcuts in fine-tuned prompt-attack defenses, with false-refusal increases up to 50% and accuracy drops up to 40%.
- Evaluated two base models and multiple defense pipelines, showing that high attack-rejection rates often arise from surface correlations rather than harmful intent.

Agentified Assessment of Logical Reasoning Agents (ICLR 2026 AIWILD Workshop)

- Designed a reproducible assessor-agent framework with structured failure tracking and standardized agent-to-agent interfaces; achieved 86.70% on cleaned FOLIO versus 73.89% for chain-of-thought reasoning.
- Implemented an auto-formalization agent that translates natural-language logic into executable Z3Py programs under auditable execution budgets.

Draft-and-Prune: Reliable Auto-formalization for Logical Reasoning (arXiv 2026)

- Proposed an inference-time framework that drafts diverse plans, prunes contradictory or ambiguous formalizations, and aggregates verified paths; reached 78.43% on AR-LSAT and 100% on PrOntoQA and LogicalDeduction.
- Improved semantic reliability without additional supervision by combining multi-path generation with solver-based verification and majority voting.

SKILLS

Python, C, C++, PyTorch, vLLM, SGLang

SERVICE

Reviewer: ICASSP 2025, ICLR 2025, IJCNN 2025, ICLR 2026, ICASSP 2026, Neurips 2026